Formal Verification of BLR PCP

yuxi-zheng

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Chapter 1

BLR for general finite fields

1.1 BLR over General Finite Fields

Throughout this section, let $q = p^s$ for a prime p and $s \in \mathbb{N}$ with $s \geq 1$. All vectors are in $\mathbb{F}_q^n$, and all expectations are uniform over the indicated finite sets. Let $\omega_p = e^{2\pi i/p}$.

Definition 1.1.1 (Field trace). *The field trace* $\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$ *is*

$$\mathrm{Tr}(z) := \sum_{i=0}^{s-1} z^{p^i}.$$

Definition 1.1.2 (Additive characters). *For* $\alpha \in \mathbb{F}_q^n$*, define the additive character* $\chi_\alpha : \mathbb{F}_q^n \to \mathbb{C}$ *by*

$$\chi_\alpha(x) := \omega_p^{\mathrm{Tr}(\langle \alpha, x \rangle)},$$

where $\langle \alpha, x \rangle = \sum_{i=1}^n \alpha_i x_i \in \mathbb{F}_q$*.*

Definition 1.1.3 (Inner product and Fourier coefficient). *For functions* $h_1, h_2 : \mathbb{F}_q^n \to \mathbb{C}$*, define*

$$\langle h_1, h_2 \rangle := \mathbb{E}_{x \in \mathbb{F}_q^n} \left[h_1(x) \overline{h_2(x)} \right].$$

For $h : \mathbb{F}_q^n \to \mathbb{C}$ *and* $\alpha \in \mathbb{F}_q^n$*, define*

$$\hat{h}(\alpha) := \langle h, \chi_\alpha \rangle = \mathbb{E}_{x \in \mathbb{F}_q^n} \left[h(x) \overline{\chi_\alpha(x)} \right].$$

Lemma 1.1.4 (Character orthogonality). *For every* $\alpha, \beta \in \mathbb{F}_q^n$*,*

$$\mathbb{E}_{x \in \mathbb{F}_q^n} \left[\chi_\alpha(x) \overline{\chi_\beta(x)} \right] = \begin{cases} 1 & \text{if } \alpha = \beta, \\ 0 & \text{if } \alpha \neq \beta. \end{cases}$$

Equivalently, $\mathbb{E}_x[\chi_\alpha(x)] = 0$ *for* $\alpha \neq 0$ *and equals* 1 *for* $\alpha = 0$*.*

Proof. First observe that for every $\alpha, \beta \in \mathbb{F}_q^n$ and every $x \in \mathbb{F}_q^n$,

$$\chi_\alpha(x)\overline{\chi_\beta(x)} = \omega_p^{\mathrm{Tr}(\langle \alpha, x \rangle)} \omega_p^{-\mathrm{Tr}(\langle \beta, x \rangle)} = \omega_p^{\mathrm{Tr}(\langle \alpha - \beta, x \rangle)} = \chi_{\alpha - \beta}(x),$$

where we used the $\mathbb{F}_p$-linearity of the trace map.

If $\alpha = \beta$, then $\alpha - \beta = 0$, so $\chi_{\alpha-\beta}(x) = 1$ for every $x \in \mathbb{F}_q^n$. Hence

$$\mathbb{E}_{x\in\mathbb{F}_q^n}\left[\chi_\alpha(x)\overline{\chi_\beta(x)}\right] = \mathbb{E}_{x\in\mathbb{F}_q^n}[1] = 1.$$

Now suppose $\alpha \neq \beta$. Let $\gamma := \alpha - \beta$. Then $\gamma \neq 0$, so there exists an index j such that $\gamma_j \neq 0$. Fix all coordinates of x except x_j. For the fixed remaining coordinates, the map

$$x_j \mapsto \langle \gamma, x \rangle$$

has the form

$$x_j \mapsto \gamma_j x_j + c$$

for some constant $c \in \mathbb{F}_q$. Since $\gamma_j \neq 0$, multiplication by γ_j is a bijection on $\mathbb{F}_q$, and therefore the affine map $x_j \mapsto \gamma_j x_j + c$ is also a bijection on $\mathbb{F}_q$. Thus, as x_j ranges uniformly over $\mathbb{F}_q$, the value $\langle \gamma, x \rangle$ also ranges uniformly over $\mathbb{F}_q$. Averaging first over x_j and then over the remaining coordinates gives

$$\mathbb{E}_{x\in\mathbb{F}_q^n}\left[\chi_\gamma(x)\right] = \mathbb{E}_{t\in\mathbb{F}_q}\left[\omega_p^{\mathrm{Tr}(t)}\right].$$

It remains to show that this last average is zero. The trace map $\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$ is a nonzero $\mathbb{F}_p$-linear map, so every fiber of Tr has the same cardinality. Since $\left|\mathbb{F}_q\right| = q$ and $\left|\mathbb{F}_p\right| = p$, each fiber has cardinality q/p. Therefore

$$\begin{aligned}
\mathbb{E}_{t\in\mathbb{F}_q}\left[\omega_p^{\mathrm{Tr}(t)}\right] &= \frac{1}{q}\sum_{t\in\mathbb{F}_q}\omega_p^{\mathrm{Tr}(t)} \\
&= \frac{1}{q}\sum_{r\in\mathbb{F}_p}\sum_{\substack{t\in\mathbb{F}_q\\ \mathrm{Tr}(t)=r}}\omega_p^r \\
&= \frac{1}{q}\sum_{r\in\mathbb{F}_p}\frac{q}{p}\omega_p^r \\
&= \frac{1}{p}\sum_{r\in\mathbb{F}_p}\omega_p^r \\
&= 0,
\end{aligned}$$

because the sum of all p-th roots of unity is zero. Hence, if $\alpha \neq \beta$, then

$$\mathbb{E}_{x\in\mathbb{F}_q^n}\left[\chi_\alpha(x)\overline{\chi_\beta(x)}\right] = 0.$$

Combining the two cases proves the stated orthogonality relation. The equivalent formulation follows by taking $\beta = 0$. ☐

Lemma 1.1.5 (Fourier inversion and Parseval)**.** *For every function $h : \mathbb{F}_q^n \to \mathbb{C}$,*

$$h(x) = \sum_{\alpha\in\mathbb{F}_q^n}\hat{h}(\alpha)\chi_\alpha(x) \quad \text{for every } x \in \mathbb{F}_q^n,$$

and

$$\sum_{\alpha\in\mathbb{F}_q^n}\left|\hat{h}(\alpha)\right|^2 = \mathbb{E}_{x\in\mathbb{F}_q^n}\left[|h(x)|^2\right].$$

In particular, if $|h(x)| = 1$ for every x, then $\sum_{\alpha\in\mathbb{F}_q^n}\left|\hat{h}(\alpha)\right|^2 = 1$.

Proof. By Lemma 1.1.4, the additive characters

$$\{\chi_\alpha : \alpha \in \mathbb{F}_q^n\}$$

are orthonormal with respect to the inner product

$$\langle h_1, h_2 \rangle = \mathbb{E}_{x \in \mathbb{F}_q^n} \left[h_1(x) \overline{h_2(x)} \right].$$

Moreover, these characters span the vector space of all complex-valued functions on $\mathbb{F}_q^n$. Indeed, there are exactly q^n characters, and the space of functions $\mathbb{F}_q^n \to \mathbb{C}$ has dimension q^n. Since the characters are nonzero and pairwise orthogonal, they are linearly independent. Hence they form an orthonormal basis.

Therefore every function $h : \mathbb{F}_q^n \to \mathbb{C}$ has a unique expansion in this basis:

$$h = \sum_{\alpha \in \mathbb{F}_q^n} c_\alpha \chi_\alpha$$

for some coefficients $c_\alpha \in \mathbb{C}$. Taking the inner product of both sides with χ_β, we obtain

$$\langle h, \chi_\beta \rangle = \sum_{\alpha \in \mathbb{F}_q^n} c_\alpha \langle \chi_\alpha, \chi_\beta \rangle.$$

By orthonormality, all terms in the sum vanish except the term $\alpha = \beta$, and $\langle \chi_\beta, \chi_\beta \rangle = 1$. Hence

$$\langle h, \chi_\beta \rangle = c_\beta.$$

By definition of the Fourier coefficient,

$$\hat{h}(\beta) = \langle h, \chi_\beta \rangle.$$

Thus $c_\beta = \hat{h}(\beta)$ for every $\beta \in \mathbb{F}_q^n$. Therefore

$$h = \sum_{\alpha \in \mathbb{F}_q^n} \hat{h}(\alpha) \chi_\alpha.$$

Evaluating this identity at $x \in \mathbb{F}_q^n$ gives the Fourier inversion formula

$$h(x) = \sum_{\alpha \in \mathbb{F}_q^n} \hat{h}(\alpha) \chi_\alpha(x).$$

It remains to prove Parseval's identity. Using the Fourier expansion and orthonormality, we compute

$$\begin{aligned}
\mathbb{E}_x |h(x)|^2 &= \langle h, h \rangle \\
&= \left\langle \sum_{\alpha \in \mathbb{F}_q^n} \hat{h}(\alpha) \chi_\alpha, \sum_{\beta \in \mathbb{F}_q^n} \hat{h}(\beta) \chi_\beta \right\rangle \\
&= \sum_{\alpha, \beta \in \mathbb{F}_q^n} \hat{h}(\alpha) \overline{\hat{h}(\beta)} \langle \chi_\alpha, \chi_\beta \rangle.
\end{aligned}$$

Again, by orthonormality, $\langle \chi_\alpha, \chi_\beta \rangle = 0$ when $\alpha \neq \beta$, and $\langle \chi_\alpha, \chi_\alpha \rangle = 1$. Hence the double sum reduces to

$$\mathbb{E}_x |h(x)|^2 = \sum_{\alpha \in \mathbb{F}_q^n} \hat{h}(\alpha) \overline{\hat{h}(\alpha)} = \sum_{\alpha \in \mathbb{F}_q^n} |\hat{h}(\alpha)|^2.$$

This proves Parseval's identity.

Finally, if $|h(x)| = 1$ for every $x \in \mathbb{F}_q^n$, then

$$\mathbb{E}_x |h(x)|^2 = \mathbb{E}_x 1 = 1.$$

Therefore Parseval gives

$$\sum_{\alpha \in \mathbb{F}_q^n} |\hat{h}(\alpha)|^2 = 1.$$

$\square$

Definition 1.1.6 (Fourier expansion of a finite-field-valued function)**.** *Let $f : \mathbb{F}_q^n \to \mathbb{F}_q$. For each $c \in \mathbb{F}_q^\times$, define*

$$\varphi_c(x) := \omega_p^{\mathrm{Tr}(cf(x))}.$$

The Fourier expansion of f is the family

$$\{\varphi_c\}_{c \in \mathbb{F}_q^\times}.$$

Lemma 1.1.7 (Character-sum indicator)**.** *For every $z \in \mathbb{F}_q$,*

$$\mathbb{1}[z = 0] = \frac{1}{q} \sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(cz)}.$$

Consequently, for every $u, v \in \mathbb{F}_q$,

$$\mathbb{1}[u = v] = \frac{1}{q} \sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(c(u-v))}.$$

Proof. Let

$$\psi(z) := \omega_p^{\mathrm{Tr}(z)}$$

be the standard additive character of $\mathbb{F}_q$. We prove that, for every $t \in \mathbb{F}_q$,

$$\mathbb{1}[t = 0] = \frac{1}{q} \sum_{c \in \mathbb{F}_q} \psi(ct).$$

The desired statement follows by taking $t = u - v$.

If $t = 0$, then $ct = 0$ for every $c \in \mathbb{F}_q$, so

$$\frac{1}{q} \sum_{c \in \mathbb{F}_q} \psi(ct) = \frac{1}{q} \sum_{c \in \mathbb{F}_q} 1 = 1.$$

Now suppose $t \neq 0$. Then multiplication by t is a bijection $\mathbb{F}_q \to \mathbb{F}_q$, so

$$\sum_{c \in \mathbb{F}_q} \psi(ct) = \sum_{z \in \mathbb{F}_q} \psi(z).$$

We claim that the latter sum is 0. Let

$$S := \sum_{z \in \mathbb{F}_q} \psi(z).$$

Since $\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$ is a nonzero $\mathbb{F}_p$-linear map, it is surjective. Hence there exists $a \in \mathbb{F}_q$ such that $\mathrm{Tr}(a) = 1$. Using the bijection $z \mapsto z + a$, we get

$$S = \sum_{z\in\mathbb{F}_q} \psi(z+a) = \sum_{z\in\mathbb{F}_q} \omega_p^{\mathrm{Tr}(z+a)} = \sum_{z\in\mathbb{F}_q} \omega_p^{\mathrm{Tr}(z)}\omega_p^{\mathrm{Tr}(a)} = \omega_p S.$$

Since $\omega_p \neq 1$, this implies $S = 0$. Therefore, when $t \neq 0$,

$$\frac{1}{q}\sum_{c\in\mathbb{F}_q} \psi(ct) = 0.$$

Combining the two cases gives

$$\mathbb{1}[t = 0] = \frac{1}{q}\sum_{c\in\mathbb{F}_q} \omega_p^{\mathrm{Tr}(ct)}.$$

Substituting $t = u - v$, we obtain

$$\mathbb{1}[u = v] = \frac{1}{q}\sum_{c\in\mathbb{F}_q} \omega_p^{\mathrm{Tr}(c(u-v))}.$$

$\square$

Definition 1.1.8 (Relative Hamming distance)**.** *For $f, g : \mathbb{F}_q^n \to \mathbb{F}_q$, define*

$$\delta(f, g) := \Pr_{x\in\mathbb{F}_q^n}\left[f(x) \neq g(x)\right].$$

Definition 1.1.9 (Linear functions)**.** *For $\alpha \in \mathbb{F}_q^n$, define $\ell_\alpha : \mathbb{F}_q^n \to \mathbb{F}_q$ by*

$$\ell_\alpha(x) := \langle \alpha, x\rangle.$$

The set of linear functions is

$$\mathcal{LIN} := \left\{\ell_\alpha \mid \alpha \in \mathbb{F}_q^n\right\}.$$

The distance from $f : \mathbb{F}_q^n \to \mathbb{F}_q$ to linearity is

$$\delta(f, \mathcal{LIN}) := \min_{\alpha\in\mathbb{F}_q^n} \delta(f, \ell_\alpha).$$

Lemma 1.1.10 (Well-separatedness of $\mathcal{LIN}$)**.** *For any distincit $f, g \in \mathcal{LIN}$, one has $\delta(f, g) = 1 - 1/|\mathbb{F}|$.*

Proof. Let $a, b \in \mathbb{F}^n$ be the unique vectors such that $f(x) = \langle a, x\rangle$ and $g(x) = \langle b, x\rangle$ for each x. Let $c = a - b$, which must be a non-zero vector. In particular, there exists $i \in [n]$ such that $c_i \neq 0$. Take any permutation $\sigma : [n] \to [n]$ such that $\sigma(i) = n$. Then

$$\begin{aligned}
\delta(f, g) &= 1 - \Pr_{x\sim\mathbb{F}^n}[\langle c, x\rangle = 0] \\
&= 1 - \frac{1}{|\mathbb{F}|^{n-1}} \sum_{x_1,\ldots,x_{n-1}\in\mathbb{F}} \frac{1}{|\mathbb{F}|}\sum_{x_n\in\mathbb{F}} \mathbb{1}[c_i \cdot x_n = -\sum_{j\in[n]\ i} c_j \cdot x_{\sigma(j)}] \\
&= 1 - \frac{1}{|\mathbb{F}|^{n-1}} \sum_{x_1,\ldots,x_{n-1}\in\mathbb{F}} \frac{1}{|\mathbb{F}|}\sum_{x_n\in\mathbb{F}} \mathbb{1}[x_n = -c_i^{-1} \cdot \sum_{j\in[n]\ i} c_j \cdot x_{\sigma(j)}]
\end{aligned}$$

where the last equality follows because $c_i \neq 0$. Now note that $\mathbb{1}[x_n = -c_i^{-1} \cdot \sum_{j \in [n] \setminus i} c_j \cdot x_{\sigma(j)}]$ evaluates to 1 for exactly one value of x_n. Thus we continue from above and get

$$\delta(f, g) = 1 - \frac{1}{|\mathbb{F}|^{n-1}} \sum_{x_1, \ldots, x_{n-1} \in \mathbb{F}} \frac{1}{|\mathbb{F}|} = 1 - \frac{1}{|\mathbb{F}|} .$$

□

Lemma 1.1.11 (Distance formula)**.** *Let $f, g : \mathbb{F}_q^n \to \mathbb{F}_q$. Let $\{\varphi_c\}_{c \in \mathbb{F}_q^\times}$ and $\{\gamma_c\}_{c \in \mathbb{F}_q^\times}$ be their Fourier expansions. Then*

$$\delta(f, g) = 1 - \frac{1}{q} \left(1 + \sum_{c \in \mathbb{F}_q^\times} \langle \varphi_c, \gamma_c \rangle \right) .$$

Equivalently,

$$\delta(f, g) = 1 - \frac{1}{q} \left(1 + \sum_{c \in \mathbb{F}_q^\times} \sum_{\alpha \in \mathbb{F}_q^n} \widehat{\varphi_c}(\alpha) \overline{\widehat{\gamma_c}(\alpha)} \right) .$$

Proof. By Theorem 1.1.7,

$$\Pr_x[f(x) = g(x)] = \mathbb{E}_x \left[\frac{1}{q} \sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(c(f(x) - g(x)))} \right] .$$

The term $c = 0$ contributes $1/q$. For $c \neq 0$,

$$\omega_p^{\mathrm{Tr}(c(f(x) - g(x)))} = \varphi_c(x) \overline{\gamma_c(x)}.$$

Therefore

$$\Pr_x[f(x) = g(x)] = \frac{1}{q} \left(1 + \sum_{c \in \mathbb{F}_q^\times} \langle \varphi_c, \gamma_c \rangle \right) ,$$

which gives the first formula after using $\delta(f, g) = 1 - \Pr_x[f(x) = g(x)]$. The second formula follows from Fourier inversion and Parseval. □

Lemma 1.1.12 (Fourier expansion of a linear function)**.** *Fix $\alpha \in \mathbb{F}_q^n$ and $c \in \mathbb{F}_q^\times$. Let $\lambda_c(x) := \omega_p^{\mathrm{Tr}(c\ell_\alpha(x))}$. Then*

$$\lambda_c = \chi_{c\alpha}.$$

Equivalently, for every $\beta \in \mathbb{F}_q^n$,

$$\widehat{\lambda_c}(\beta) = \begin{cases} 1 & \text{if } \beta = c\alpha, \\ 0 & \text{if } \beta \neq c\alpha. \end{cases}$$

Proof. For every $x \in \mathbb{F}_q^n$,

$$\lambda_c(x) = \omega_p^{\mathrm{Tr}(c\langle \alpha, x \rangle)} = \omega_p^{\mathrm{Tr}(\langle c\alpha, x \rangle)} = \chi_{c\alpha}(x).$$

The statement about Fourier coefficients follows from Theorem 1.1.4. □

Lemma 1.1.13 (Distance to linearity in Fourier form)**.** *Let $f : \mathbb{F}_q^n \to \mathbb{F}_q$ have Fourier expansion $\{\varphi_c\}_{c\in\mathbb{F}_q^\times}$. Then*

$$\delta(f, \mathcal{LIN}) = 1 - \frac{1}{q}\left(1 + \max_{\alpha\in\mathbb{F}_q^n} \sum_{c\in\mathbb{F}_q^\times} \widehat{\varphi_c}(c\alpha)\right).$$

Moreover, for every $\alpha \in \mathbb{F}_q^n$, the quantity

$$S_\alpha := \sum_{c\in\mathbb{F}_q^\times} \widehat{\varphi_c}(c\alpha)$$

is real and satisfies $-1 \le S_\alpha \le q-1$.

Proof. Apply Theorem 1.1.11 with $g = \ell_\alpha$ and use Theorem 1.1.12. This gives

$$\delta(f, \ell_\alpha) = 1 - \frac{1}{q}\left(1 + \sum_{c\in\mathbb{F}_q^\times} \widehat{\varphi_c}(c\alpha)\right).$$

Taking the minimum over α gives the claimed formula for $\delta(f, \mathcal{LIN})$. For fixed α, the same formula expresses S_α as $q(1-\delta(f,\ell_\alpha)) - 1$. Hence S_α is real, and since $0 \le \delta(f,\ell_\alpha) \le 1$, we get $-1 \le S_\alpha \le q-1$. □

Definition 1.1.14 (Finite-field BLR test)**.** *Given oracle access to $f : \mathbb{F}_q^n \to \mathbb{F}_q$, the finite-field BLR verifier $\mathsf{V}^f_{\mathsf{BLR}}$ samples $a, b \in \mathbb{F}_q^\times$ and $x, y \in \mathbb{F}_q^n$ uniformly and accepts iff*

$$af(x) + bf(y) = f(ax+by).$$

That is,

$$\mathsf{V}^f_{\mathsf{BLR}} = 1 \iff af(x) + bf(y) = f(ax+by).$$

Lemma 1.1.15 (Completeness)**.** *If $f \in \mathcal{LIN}$, then*

$$\Pr[\mathsf{V}^f_{\mathsf{BLR}} = 1] = 1.$$

Proof. If $f = \ell_\alpha$ for some $\alpha \in \mathbb{F}_q^n$, then for every $a, b \in \mathbb{F}_q^\times$ and $x, y \in \mathbb{F}_q^n$,

$$f(ax+by) = \langle \alpha, ax+by\rangle = a\langle\alpha, x\rangle + b\langle\alpha, y\rangle = af(x) + bf(y).$$

Thus the verifier accepts for every choice of randomness. □

Lemma 1.1.16 (Exact BLR acceptance formula)**.** *Let $f : \mathbb{F}_q^n \to \mathbb{F}_q$ have Fourier expansion $\{\varphi_c\}_{c\in\mathbb{F}_q^\times}$. For each $\alpha \in \mathbb{F}_q^n$, define*

$$S_\alpha := \sum_{c\in\mathbb{F}_q^\times} \widehat{\varphi_c}(c\alpha).$$

Then

$$\Pr[\mathsf{V}^f_{\mathsf{BLR}} = 1] = \frac{1}{q}\left(1 + \frac{1}{(q-1)^2}\sum_{\alpha\in\mathbb{F}_q^n} S_\alpha^3\right).$$

Proof. By Theorem 1.1.7, for fixed a, b, x, y,

$$\mathbb{1}[af(x) + bf(y) = f(ax + by)] = \frac{1}{q} \sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(c(af(x)+bf(y)-f(ax+by)))}.$$

Taking expectation gives

$$\Pr[\mathsf{V}^f_{\mathsf{BLR}} = 1] = \frac{1}{q} + \frac{1}{q}\mathbb{E}_{a,b,x,y}\left[\sum_{c \in \mathbb{F}_q^\times} \varphi_{ca}(x)\varphi_{cb}(y)\varphi_{-c}(ax + by)\right].$$

Expand the three complex-valued functions into their Fourier expansions:

$$\begin{aligned}
&\mathbb{E}_{a,b,x,y}\left[\sum_{c \in \mathbb{F}_q^\times} \varphi_{ca}(x)\varphi_{cb}(y)\varphi_{-c}(ax + by)\right] \\
&= \mathbb{E}_{a,b}\left[\sum_{c \in \mathbb{F}_q^\times} \sum_{\alpha,\beta,\gamma \in \mathbb{F}_q^n} \widehat{\varphi_{ca}}(\alpha)\widehat{\varphi_{cb}}(\beta)\widehat{\varphi_{-c}}(\gamma)\mathbb{E}_{x,y}\left[\chi_\alpha(x)\chi_\beta(y)\chi_\gamma(ax + by)\right]\right].
\end{aligned}$$

Since

$$\chi_\gamma(ax + by) = \chi_{a\gamma}(x)\chi_{b\gamma}(y),$$

Theorem 1.1.4 implies that the inner expectation is 1 exactly when

$$\alpha + a\gamma = 0 \quad \text{and} \quad \beta + b\gamma = 0,$$

and is 0 otherwise. Therefore the last display equals

$$\mathbb{E}_{a,b}\left[\sum_{c \in \mathbb{F}_q^\times} \sum_{\alpha \in \mathbb{F}_q^n} \widehat{\varphi_{ca}}(\alpha)\widehat{\varphi_{cb}}(a^{-1}b\alpha)\widehat{\varphi_{-c}}(-a^{-1}\alpha)\right].$$

Substitute $\alpha = ca\eta$. Then this becomes

$$\mathbb{E}_{a,b}\left[\sum_{c \in \mathbb{F}_q^\times} \sum_{\eta \in \mathbb{F}_q^n} \widehat{\varphi_{ca}}(ca\eta)\widehat{\varphi_{cb}}(cb\eta)\widehat{\varphi_{-c}}(-c\eta)\right].$$

As a, b, c range over $\mathbb{F}_q^\times$, so do ca, cb, and $-c$. Hence this equals

$$\frac{1}{(q-1)^2} \sum_{\eta \in \mathbb{F}_q^n} \left(\sum_{d \in \mathbb{F}_q^\times} \widehat{\varphi_d}(d\eta)\right)^3 = \frac{1}{(q-1)^2} \sum_{\eta \in \mathbb{F}_q^n} S_\eta^3.$$

Substituting into the expression for $\Pr[\mathsf{V}^f_{\mathsf{BLR}} = 1]$ proves the claim. □

Lemma 1.1.17 (Second moment bound)**.** *With S_α as in Theorem 1.1.16,*

$$\sum_{\alpha \in \mathbb{F}_q^n} S_\alpha^2 \le (q-1)^2.$$

Proof. Expanding the square gives

$$\sum_{\alpha \in \mathbb{F}_q^n} S_\alpha^2 = \sum_{a,b \in \mathbb{F}_q^\times} \sum_{\alpha \in \mathbb{F}_q^n} \widehat{\varphi_a}(a\alpha)\widehat{\varphi_b}(b\alpha).$$

For fixed $a, b \in \mathbb{F}_q^\times$, Cauchy–Schwarz and Theorem 1.1.5 give

$$\begin{aligned} \left| \sum_{\alpha \in \mathbb{F}_q^n} \widehat{\varphi_a}(a\alpha)\widehat{\varphi_b}(b\alpha) \right| &\leq \left(\sum_{\alpha \in \mathbb{F}_q^n} |\widehat{\varphi_a}(a\alpha)|^2 \right)^{1/2} \left(\sum_{\alpha \in \mathbb{F}_q^n} |\widehat{\varphi_b}(b\alpha)|^2 \right)^{1/2} \\ &= 1. \end{aligned}$$

The equality uses that multiplication by a and by b permutes $\mathbb{F}_q^n$, and that $|\varphi_a(x)| = |\varphi_b(x)| = 1$ for every x. Summing over the $(q-1)^2$ choices of (a, b) gives the result. $\square$

Theorem 1.1.18 (Soundness of the finite-field BLR test)**.** *For every function* $f : \mathbb{F}_q^n \to \mathbb{F}_q$*,*

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 0] \geq \delta(f, \mathcal{LIN}).$$

Equivalently,

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 1] \leq 1 - \delta(f, \mathcal{LIN}).$$

Proof. Let S_α be as in Theorem 1.1.16, and let

$$M := \max_{\alpha \in \mathbb{F}_q^n} S_\alpha.$$

By Theorem 1.1.13,

$$1 - \delta(f, \mathcal{LIN}) = \frac{1}{q}(1 + M).$$

By Theorem 1.1.13, each S_α is real and $S_\alpha \leq M$. Therefore $S_\alpha^3 \leq M S_\alpha^2$ for every α, because $S_\alpha^2 \geq 0$. Using Theorem 1.1.16 and Theorem 1.1.17,

$$\begin{aligned} \Pr[\mathsf{V}_{\mathsf{BLR}}^f = 1] &= \frac{1}{q}\left(1 + \frac{1}{(q-1)^2} \sum_{\alpha \in \mathbb{F}_q^n} S_\alpha^3\right) \\ &\leq \frac{1}{q}\left(1 + \frac{M}{(q-1)^2} \sum_{\alpha \in \mathbb{F}_q^n} S_\alpha^2\right) \\ &\leq \frac{1}{q}(1 + M) \\ &= 1 - \delta(f, \mathcal{LIN}). \end{aligned}$$

Taking complements gives $\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 0] \geq \delta(f, \mathcal{LIN})$. $\square$

Chapter 2

Gowers Hatami theorem

2.1 Preliminaries

Definition 2.1.1 (σ inner product.). *The σ inner product between matrices A and B is defined as*

$$\langle A, B\rangle_\sigma = \mathrm{Tr}(A^* B\sigma), \tag{2.1}$$

where $\sigma \geq 0$ and A^* denotes conjugate-transpose.

Definition 2.1.2 ((ϵ, σ)-representation.). *Given a finite group G, an integer $d \geq 1, \varepsilon \geq 0$, and a d-dimensional positive semidefinite matrix σ with trace 1 , an (ε, σ)-representation of G is a map $f : G \to U_d(\mathbb{C})$, to the $d \times d$ unitary matrices, such that*

$$\mathbb{E}_{x,y\in G}\Re\left(\langle f(x)^* f(y), f\left(x^{-1}y\right)\rangle_\sigma\right) \geq 1 - \varepsilon, \tag{2.2}$$

where $\Re$ denotes taking real part of the inner product.

Proposition 2.1.3. *Let G be a finite group, and let $f, f_2 : G \to U(n)$ be functions into the unitary matrices. Then*

$$\mathbb{E}_{x\in G}\|f(x) - f_2(x)\|_\sigma^2 \leq 2\varepsilon \quad \Longleftrightarrow \quad \mathbb{E}_{x\in G}\Re(\langle f(x), f_2(x)\rangle_\sigma) \geq 1 - \varepsilon.$$

Proof. For unitary matrices A, B, we have

$$\|A - B\|_\sigma^2 = 2 - 2\Re\langle A, B\rangle_\sigma. \tag{2.3}$$

Since $\mathbb{E}$ is linear, this proves the proposition. $\square$

TODO:

- Relation to classical BLR test

2.2 Representation theory

Definition 2.2.1 (Unitary Representation). *A* unitary representation *of G is a representation to unitary matrices $\rho : G \to U(V)$.*

Definition 2.2.2 (Regular representation)**.** *Let $R : G \to \mathbb{C}[G]$ be a regular representation where $\{|g\rangle, g \in G\}$ forms a basis of $\mathbb{C}[G]$. The representation is given by*

$$R(x)|g\rangle = |xg\rangle. \tag{2.4}$$

It is possible to decompose any representation of a finite group into direct sums over irreps.

Theorem 2.2.3 (Maschke's theorem)**.** *Let G be a finite group and let $\rho : G \to \mathrm{GL}(V)$ be a finite-dimensional representation over $\mathbb{C}$. Then ρ is completely reducible. In particular, there exists a decomposition*

$$\rho \cong \bigoplus_m r_m \, \rho_m, \tag{2.5}$$

where $\{\rho_m\}$ is a set of inequivalent irreducible representations of G, and $r_m \in \mathbb{Z}^+$ are the multiplicities.

Proof sketch. Maschke's theorem is partially formalized in `Mathlib.RepresentationTheory.Maschke` but incomplete. Jonathan's team is formalizing this. □

Proposition 2.2.4 (Decomposition of the regular representation)**.** *Let $R : G \to \mathrm{GL}(\mathbb{C}[G])$ be the regular representation of a finite group G. Then*

$$R \cong \bigoplus_{\rho \in \widehat{G}} d_\rho \, \rho, \tag{2.6}$$

where $\widehat{G}$ is a complete set of inequivalent irreducible representations of G, and $d_\rho = \dim(V_\rho)$ is the dimension of ρ.

Proof sketch. This can be proven using Maschke's theorem in 2.2.3 to decompose into irreps with multiplicity. Then use character orthogonality theorem in mathlib `FDRep.char_orthonormal` to show the multiplicity equals irrep dimension. A more formal statement is given by the Peter-Weyl theorem. □

Proposition 2.2.5 (Character of regular representation.)**.** *Let G be a finite group and $\sum_\rho$ denotes summing over the complete set of inequivalent irreps $\hat{G}$*

$$\sum_{\rho \in \hat{G}} d_\rho \, \mathrm{Tr}(\rho(x)) = |G| \delta_{x,e}. \tag{2.7}$$

Proof sketch. The proof uses decomposition of regular representation and definition of character. □

Definition 2.2.6 (Group fourier transform)**.** *Let $f : G \to U_d(\mathbb{C})$ be a map and $\rho : G \to U_{d_\rho}(\mathbb{C})$ be a unitary irrep. The fourier transform of f at the representation ρ is denoted by $\hat{f}$:*

$$\hat{f}(\rho) = \mathbb{E}_{x \in G} f(x) \otimes \overline{\rho(x)}, \tag{2.8}$$

where $\overline{\rho(x)}$ denotes complex conjugate of $\rho(x)$.

2.3 Gowers Hatami Theorem

Theorem 2.3.1 (Gowers-Hatami)**.** *Let G be a finite group, $\varepsilon \geq 0$, and $f : G \to U_d(\mathbb{C})$ an (ε, σ)-representation of G. Then there exists a $d' \geq d$, an isometry $V : \mathbb{C}^d \to \mathbb{C}^{d'}$, and a representation $g : G \to U_{d'}(\mathbb{C})$ such that*

$$\mathbb{E}_{x\in G} \left\| f(x) - V^* g(x) V \right\|_\sigma^2 \leq 2\varepsilon. \tag{2.9}$$

Proof sketch. The full proof can be found in the lecture notes Theorem 2.3 [?]. In particular, the proof is constructive. The isometry is defined as $V : \mathbb{C}^d \to \mathbb{C}^d \otimes \left(\oplus_\rho \mathbb{C}^{d_\rho} \otimes \mathbb{C}^{d_\rho}\right)$ by

$$Vu = \bigoplus_\rho d_\rho^{1/2} \sum_{i=1}^{d_\rho} \left(\hat{f}(\rho)\,(u \otimes e_i)\right) \otimes e_i. \tag{2.10}$$

And the exact representation is defined as

$$g(x) = \bigoplus_\rho \left(I_d \otimes I_{d_\rho} \otimes \rho(x)\right). \tag{2.11}$$

The proof requires the following two ingredients:

1. V is an isometry:
$$V^* V = \mathbb{I}_d \tag{2.12}$$

2. The isometry "averages" over group multiplication. For any $x \in G$,
$$V^* g(x) V = \mathbb{E}_z f(z)^* f(zx). \tag{2.13}$$

Both ingredients follow from the key identity over the regular representation in Eq. (2.7). □

Next, we will give a different proof that does not directly decompose into irreps but instead uses the action of the regular representation.

Lemma 2.3.2 (Gowers-Hatami-Correlation)**.** *Let G be a finite group, $\varepsilon \geq 0$, and $\rho : G \to U(\mathcal{H})$ an (ε, σ)-representation of G on space $\mathcal{H}$. Then there exists an isometry $V : \mathcal{H} \to \mathcal{H}'$ to a different space $\mathcal{H}'$, and a representation $\rho' : G \to \mathcal{H}'$ such that the average correlation between ρ and ρ' is:*

$$\mathbb{E}_{x\in G} \Re \langle f(x), V^* \rho'(x) V\rangle_\sigma \geq 1 - \varepsilon, \tag{2.14}$$

Proof. Here $\mathcal{H}' = L(G, \mathcal{H})$ is the space of functions $f : G \to \mathcal{H}$.

Let the isometry be defined by the action of the approximate representation on a state $u \in \mathcal{H}$ as

$$\begin{aligned} V :&\quad \mathcal{H} \to L(G, \mathcal{H}), && (2.15)\\ V(u) :&\quad G \to \mathcal{H}, \quad x \mapsto \rho(x)(u), \forall x \in G. && (2.16) \end{aligned}$$

The adjoint map V^* is defined as the unique map such that for all $u \in \mathcal{H}$ and $g \in L(G, \mathcal{H})$,

$$\langle V(u), g\rangle_{L(G,\mathcal{H})} = \langle u, V^*(g)\rangle_{\mathcal{H}}. \tag{2.17}$$

In particular, V^* is explicitly given by the group average

$$\begin{aligned} V^* :&\quad L(G, \mathcal{H}) \to \mathcal{H}, && (2.18)\\ V^*(f) &\quad = \mathbb{E}_{x\in G} \left[\rho^*(x) f(x)\right]. && (2.19) \end{aligned}$$

The exact representation is given by the right regular representation on $L(G, \mathcal{H})$:

$$\begin{aligned} \rho' : \quad & G \to \mathrm{End}(L(G,\mathcal{H})), && (2.20)\\ (\rho'(x)f)(y) \quad & = f(yx), \quad \forall x, y \in G. && (2.21) \end{aligned}$$

V is an isometry because

$$V^*V(u) = \mathbb{E}_{x\in G}\left[\rho^*(x)\rho(x)(u)\right] = u. \tag{2.22}$$

The isometry "averages" over group multiplication because

$$\begin{aligned} V^*\rho'(x)V(u) &= \mathbb{E}_{y\in G}\left[\rho^*(y)(\rho'(x)\rho(y))(u)\right] \qquad (2.23)\\ &= \mathbb{E}_{y\in G}\left[\rho^*(y)(\rho(yx)(u))\right]. \end{aligned}$$

Finally, using the isometry average property in Eq. 2.23 and the definition of (ε,σ)-representation, we have

$$\begin{aligned} & \mathbb{E}_{x\in G}\mathfrak{R}\langle \rho(x), V^*\rho'(x)V\rangle_\sigma, \qquad (2.24)\\ = \; & \mathbb{E}_{x,y\in G}\mathfrak{R}\langle \rho(x), \rho^*(y)\rho(yx)\rangle_\sigma,\\ \geq \; & 1-\varepsilon. \end{aligned}$$

□

Theorem 2.3.3 (Gowers-Hatami-2)**.** *Let G be a finite group, $\varepsilon \geq 0$, and $\rho : G \to U(\mathcal{H})$ an (ε,σ)-representation of G on space $\mathcal{H} = \mathbb{C}^d$. Then there exists an isometry $V : \mathcal{H} \to \mathcal{H}'$ to a different space $\mathcal{H}'$, and a representation $\rho' : G \to \mathcal{H}'$ such that*

$$\mathbb{E}_{x\in G}\left\|\rho(x) - V^*\rho'(x)V\right\|_\sigma^2 \leq 2\varepsilon. \tag{2.25}$$

Proof. A different proof of the Gowers-Hatami theorem is given in M. de la Salle's notes [**?**].

The rest of the proof follows by expanding the σ-norm, bounding the norm of $\mathbb{E}_{x\in G}V^*\rho'(x)V$ and applying the definition of (ε,σ)-representation. In particular, we have

$$\begin{aligned} & \mathbb{E}_{x\in G}\left\|\rho(x) - V^*\rho'(x)V\right\|_\sigma^2 \qquad (2.26)\\ = \; & \mathbb{E}_{x\in G}\|\rho(x)\|_\sigma^2 + \mathbb{E}_{x\in G}\|V^*\rho'(x)V\|_\sigma^2 - 2\mathbb{E}_{x\in G}\mathfrak{R}\langle\rho(x), V^*\rho'(x)V\rangle_\sigma \end{aligned}$$

Because ρ is unitary, $\mathbb{E}_{x\in G}\|\rho(x)\|_\sigma^2 = 1$. For the second term, since ρ' is unitary and V is an isometry,

$$\begin{aligned} & \mathbb{E}_{x\in G}\|V^*\rho'(x)V\|_\sigma^2 \qquad (2.27)\\ = \; & \mathbb{E}_{x\in G}\mathrm{Tr}\left[V^*\rho'^*(x)VV^*\rho'(x)V\sigma\right]\\ = \; & \mathbb{E}_{x,y,z\in G}\mathrm{Tr}\left[\rho^*(yx)\rho(y)\rho^*(z)\rho(zx)\sigma\right] \end{aligned}$$

Because ρ is unitary the product $U(x,y,z) := \rho^*(yx)\rho(y)\rho^*(z)\rho(zx)$ is again a unitary operator, the above expression is upper bounded by Holder's inequality as

$$\begin{aligned} & \mathbb{E}_{x,y,z\in G}\mathrm{Tr}\left[U(x,y,z)\sigma\right] \qquad (2.28)\\ \leq \; & \mathbb{E}_{x,y,z\in G}\|U(x,y,z)\|_\infty\|\sigma\|_1 = 1. \end{aligned}$$

Therefore using the bounds in Eq. 2.28 and Eq. 2.24 by applying Lemma 2.3.2, we have

$$\mathbb{E}_{x\in G}\left\|\rho(x) - V^*\rho'(x)V\right\|_\sigma^2 \leq 2 - 2(1-\varepsilon) = 2\varepsilon. \tag{2.29}$$

□

2.4 Gowers-Hatami for classical BLR

In classical BLR, the mapping is from a field $\mathbb{F}_2^n$. First let us relate the BLR test to the notion of approximate representation.

Proposition 2.4.1 (BLR test induces an approximate representation). *Let $f : \mathbb{Z}_2^n \to \mathbb{Z}_2$ satisfy*

$$\Pr_{x,y}[f(x) + f(y) = f(x + y)] \geq 1 - \epsilon.$$

Define $F : \mathbb{Z}_2^n \to \{\pm 1\}$ by $F(x) = (-1)^{f(x)}$. Then

$$\mathbb{E}_{x,y}\big|F(x)F(y) - F(x + y)\big|^2 \leq 4\epsilon,$$

so F is an $(2\epsilon, 1)$-approximate representation.

Proof. First observe that for all $x, y \in G$,

$$F(x)F(y) = (-1)^{f(x)+f(y)}.$$

Hence

$$F(x)F(y) = F(x + y) \quad \iff \quad f(x) + f(y) = f(x + y) \pmod 2.$$

Define the indicator of BLR success:

$$\mathbf{1}_{\text{good}}(x, y) := \mathbf{1}[f(x) + f(y) = f(x + y)].$$

Then

$$\Pr_{x,y}[\mathbf{1}_{\text{good}} = 1] \geq 1 - \epsilon.$$

Now compute the squared error. Since $F(x) \in \{\pm 1\}$,

$$|F(x)F(y) - F(x + y)|^2 = \begin{cases} 0 & \text{if } F(x)F(y) = F(x + y), \\ 4 & \text{otherwise.} \end{cases}$$

Taking expectation over (x, y),

$$\begin{aligned} \mathbb{E}_{x,y}|F(x)F(y) - F(x + y)|^2 &= 4 \Pr_{x,y}[F(x)F(y) \neq F(x + y)] \\ &= 4 \Pr_{x,y}[f(x) + f(y) \neq f(x + y)] \\ &\leq 4\epsilon. \end{aligned}$$

□

The Gowers-Hatami theorem can be applied to show that if a function $f : \mathbb{Z}_2^n \to \mathbb{F}_2$ passes the BLR linearity test with high probability, then there exists a linear function $\ell : \mathbb{F}_2^n \to \mathbb{F}_2$ that is close to f in terms of Hamming distance $\delta(f, \ell)$ defined in Def. 1.1.8. To do this, we first need to apply the Gowers-Hatami theorem to the approximate representation $F(x) = (-1)^{f(x)}$ given by Proposition 2.4.1.

Corollary 2.4.2 (Gowers-Hatami Correlation for BLR). *Let $F : \mathbb{Z}_2^n \to U(1)$ be an (ϵ, σ) approximate representation. Then there exists a unit vector v and an exact representation G such that*

$$\mathbb{E}_{x\in\mathbb{Z}_2^n}\Re\langle F(x), \langle v, G(x)v\rangle\rangle \geq 1 - \epsilon. \tag{2.30}$$

Next, we would like to conclude from the above corollary that there exists a linear function $\ell : \mathbb{Z}_2^n \to \mathbb{Z}_2$ such that $\Pr_x[f(x) \neq \ell(x)] \leq \epsilon/2$. To do this, we need to use a fact that about the regular representation of $\mathbb{Z}_2^n$.

Fact 2.4.3 (Regular representation of $\mathbb{Z}_2^n$)**.** *Let $\rho(a)$ act on $L = \{f : \mathbb{Z}_2^n \to \mathbb{C}\}$ by $(\rho(a)f)(x) = f(x+a)$. Then in the Fourier basis $\{|S\rangle\}$, we have*

$$\rho(a) = \sum_{S \subseteq [n]} \chi_S(a)|S\rangle\langle S|.$$

Lemma 2.4.4 (Gowers-Hatami implies near linearity)**.** *Let $F : \mathbb{Z}_2^n \to \mathbb{C}$ be an $(\epsilon, 1)$ approximate representation of $\mathbb{Z}_2^n$ from the BLR test such that $F(x) = (-1)^{f(x)}$ for some function $f : \mathbb{Z}_2^n \to \mathbb{Z}_2$. Then there exists a linear function $\ell : \mathbb{F}_2^n \to \mathbb{F}_2$ such that the distance between f and ℓ is bounded*

$$\delta(f, \ell) \leq \epsilon/2. \tag{2.31}$$

Proof. By Corollary 2.4.2, there exists a representation G and a unit vector v such that

$$\mathbb{E}_x \mathfrak{R}\langle F(x), \langle v, G(x)v\rangle\rangle \geq 1 - \epsilon.$$

Using the diagonal form of $G(x)$ in the character basis,

$$G(x) = \sum_S \chi_S(x)|S\rangle\langle S|.$$

The inner product condition can be rewritten as

$$\sum_S p_S \mathbb{E}_x \mathfrak{R}\langle F(x), \chi_S(x)\rangle \geq 1 - \epsilon,$$

where $p_S := |\langle v|S\rangle|^2$ and $\sum_S p_S = 1$. Therefore,

$$\sum_S p_S \widehat{F}(S) \geq 1 - \epsilon,$$

where $\widehat{F}(S) := \mathbb{E}_x[F(x)\chi_S(x)]$ is the real-valued Fourier coefficient of F at S.

Since $(p_S)_S$ is a probability distribution, there exists some S^* such that

$$\widehat{F}(S^*) \geq 1 - \epsilon.$$

This means that there is a large Fourier coefficient at S^*. Define $\ell(x) = x \cdot S^*$. Then $\chi_{S^*}(x) = (-1)^{\ell(x)}$, and

$$\mathbb{E}_x[F(x)\chi_{S^*}(x)] = 1 - 2\Pr_x[f(x) \neq \ell(x)].$$

Hence

$$1 - 2\delta(f, \ell) \geq 1 - \epsilon,$$

which implies

$$\delta(f, \ell) \leq \epsilon/2.$$

□

Proposition 2.4.5 (Equivalence for BLR test)**.** *Let $\Pr[V_{BLR}^f = 0]$ denote the probability that the BLR verifier rejects a function $f : \mathbb{F}_2^n \to \mathbb{F}_2$. Let $\delta(f, \mathcal{LIN})$ denote the distance between f and the closest linear function in $\mathcal{LIN}$. Then the following statements are equivalent:*

- *If* $\Pr[V_{BLR}^f = 0] \leq \epsilon$*, then* $\delta(f, \mathcal{LIN}) \leq \epsilon$*.*

- $\Pr[V_{BLR}^f = 0] \geq \delta(f, \mathcal{LIN})$*.*

Now we provide the proof of Soundness of BLR test using the Gowers-Hatami theorem.

Theorem 2.4.6 (Soundness of the Z_2 BLR)**.** *For every function* $f : \mathbb{F}_2^n \to \mathbb{F}_2$*,*

$$\Pr[V_{BLR}^f = 0] \geq \delta(f, \mathcal{LIN}).$$

Equivalently,

$$\Pr[V_{BLR}^f = 1] \leq 1 - \delta(f, \mathcal{LIN}).$$

Proof. Let $F : \mathbb{Z}_2^n \to \{-1, 1\}$ be defined by $F(x) = (-1)^{f(x)}$, as given by Proposition .2.4.1. F is an $(2\epsilon, 1)$ approximate representation of $\mathbb{Z}_2^n$. Applying Gowers-Hatami theorem by Lemma. 2.4.4, then there exists a linear function ℓ such that $\delta(f, \ell) \leq \epsilon$ so that $\delta(f, \mathcal{LIN}) \leq \epsilon$.

This proves the implication $\Pr[V_{\mathrm{BLR}}^f = 0] \leq \epsilon \Rightarrow \delta(f, \mathcal{LIN}) \leq \epsilon$, which is equivalent to

$$\Pr[V_{\mathrm{BLR}}^f = 0] \geq \delta(f, \mathcal{LIN}),$$

by Proposition 2.4.5. □

Chapter 3

Exponential-length PCP

Definition 3.0.1. *A system of m quadratic equations in n variables over a field $\mathbb{F}$ is a list of polynomials $p_1, \dots, p_m \in \mathbb{F}[x_1, \dots, x_n]$ where each p_i has total degree at most 2.*

$$\mathrm{QESAT}(\mathbb{F}, n) := \{(p_1, \dots, p_m) \mid \exists a_1, \dots, a_n \in \mathbb{F},\ \forall i \in [m],\ p_i(a_1, \dots, a_n) = 0\}.$$

For example, $(x + 1, xy + z) \in \mathrm{QESAT}(\mathbb{F}_2, 3)$.

Definition 3.0.2. *A* PCP verifier *is a probabilistic oracle Turing machine V with query access to a function $\pi : [\ell(|x|)] \to \Sigma$.*

Definition 3.0.3. *A language $L \subseteq \{0,1\}^*$ is in $\mathrm{PCP}[\varepsilon_c, \varepsilon_s, \Sigma, \ell, q, r]$ if there exists a polynomial-time PCP verifier V such that for every $x \in \{0,1\}^*$, V makes at most $q(|x|)$ queries to the proof oracle, uses at most $r(|x|)$ bits of randomness, and the following holds:*

- *Completeness: If $x \in L$, then $\exists \pi \in \Sigma^{\ell(|x|)}$, $\Pr\left[V^{\pi}(x) = 1\right] \geq 1 - \varepsilon_c$.*
- *Soundness: If $x \notin L$, then $\forall \tilde{\pi} \in \Sigma^{\ell(|x|)}$, $\Pr\left[V^{\tilde{\pi}}(x) = 1\right] \leq \varepsilon_s$.*

The following theorem is the main goal of this chapter.

Theorem 3.0.4.

$$\mathrm{QESAT}(\mathbb{F}_2, n) \in \mathrm{PCP}[\varepsilon_c = 0, \varepsilon_s = 1/2, \Sigma = \mathbb{F}_2, \ell = 2^{|x|}, q = O(1), r = |x|^{O(1)}].$$

Proof. TODO $\square$

Definition 3.0.5. *A* LPCP verifier *is a probabilistic oracle Turing machine V with query access to a linear function $\langle \pi, \cdot \rangle : \Sigma^{\ell(|x|)} \to \Sigma$.*

Definition 3.0.6. *A language $L \subseteq \{0,1\}^*$ is in $\mathrm{LPCP}[\varepsilon_c, \varepsilon_s, \Sigma, \ell, q, r]$ if there exists a polynomial-time LPCP verifier V such that for every $x \in \{0,1\}^*$, V makes at most $q(|x|)$ queries to the linear proof oracle, uses at most $r(|x|)$ bits of randomness, and the following holds:*

- *Completeness: If $x \in L$, then $\exists \pi \in \Sigma^{\ell(|x|)}$, $\Pr\left[V^{\langle \pi, \cdot \rangle}(x) = 1\right] \geq 1 - \varepsilon_c$.*
- *Soundness: If $x \notin L$, then $\forall \tilde{\pi} \in \Sigma^{\ell(|x|)}$, $\Pr\left[V^{\langle \tilde{\pi}, \cdot \rangle}(x) = 1\right] \leq \varepsilon_s$.*

3.1 Simpler BLR verifier for $\mathbb{F}_2$

Definition 3.1.1. *Let $V^f_{\mathrm{BLR}(\mathbb{F}_2,n)}$ be the verifier with oracle access to $f : \mathbb{F}_2^n \to \mathbb{F}_2$ defined as follows:*

1. *Sample $x, y \leftarrow \mathbb{F}^n$.*
2. *Accept if and only if $f(x) + f(y) = f(x + y)$.*

Theorem 3.1.2. *Let $n > 0$. For every linear function $f : \mathbb{F}_2^n \to \mathbb{F}_2$, we have*

$$\Pr\left[V^f_{\mathrm{BLR}(\mathbb{F}_2,n)} = 1\right] = 1.$$

Proof. □

Theorem 3.1.3. *Let $n > 0$ and $f : \mathbb{F}_2^n \to \mathbb{F}_2$.*

$$\Pr\left[V^f_{\mathrm{BLR}(\mathbb{F}_2,n)} = 0\right] \geq \Delta(f, \mathcal{LIN}).$$

Proof. □

3.2 Linear PCP for linear equations

Definition 3.2.1. *For a field $\mathbb{F}$ and parameters $m, n \in \mathbb{N}$, we define the language*

$$\mathrm{LINEQ}(\mathbb{F}, m, n) := \{(M, c) \in \mathbb{F}^{m\times n} \times \mathbb{F}^m \mid \exists b \in \mathbb{F}^n,\ Mb = c\}.$$

Definition 3.2.2. *Let $V^{\langle b,\cdot\rangle}_{\mathrm{LINEQ}(\mathbb{F},m,n)}(M, c)$ be the LPCP verifier defined as follows:*

1. *Sample $r \leftarrow \mathbb{F}^m$.*
2. *Query $b \in \mathbb{F}^n$ at $u := M^\top r \in \mathbb{F}^n$.*
3. *Accept if and only if $\langle b, u\rangle = \langle c, r\rangle$.*

Theorem 3.2.3.

$$\mathrm{LINEQ}(\mathbb{F}, m, n) \in \mathrm{LPCP}\left[\varepsilon_c = 0, \varepsilon_s = 1/|\mathbb{F}|, \Sigma = \mathbb{F}, \ell = |x|, q = 1, r = m \log |\mathbb{F}|\right].$$

Proof. Consider the LPCP verifier $V^{\langle b,\cdot\rangle}_{\mathrm{LINEQ}(\mathbb{F},m,n)}(M, c)$.
Completeness: If $Mb = c$, then for every $r \in \mathbb{F}^m$, we have

$$\langle b, u\rangle = b^\top(M^\top r) = (Mb)^\top r = \langle c, r\rangle.$$

Soundness: If $Mb \neq c$, then

$$\begin{aligned}\Pr_{r\leftarrow\mathbb{F}^m}\left[\langle b, u\rangle = \langle c, r\rangle\right] &= \Pr_{r\leftarrow\mathbb{F}^m}\left[\langle Mb, r\rangle = \langle c, r\rangle\right] \\ &= \Pr_{r\leftarrow\mathbb{F}^m}\left[\sum_{i=1}^m (Mb - c)_i \cdot r_i = 0\right] \leq 1/|\mathbb{F}|,\end{aligned}$$

where the last inequality follows from the Schwartz-Zippel lemma. □

3.3 Linear PCP for tensor checks

Definition 3.3.1. *For a field $\mathbb{F}$, vectors $a \in \mathbb{F}^n$ and $b \in \mathbb{F}^m$, the tensor product $a \otimes b \in \mathbb{F}^{n \times m}$ is the matrix defined by*

$$(a \otimes b)_{i,j} := a_i \cdot b_j \quad \forall i \in [n],\, j \in [m].$$

We write $\mathrm{flat}(a \otimes b) \in \mathbb{F}^{n \cdot m}$ *for the row-concatenation of $a \otimes b$.*

For our purposes, we would love to test to check if $\mathrm{flat}(a \otimes a) = b$

Definition 3.3.2. *For a field $\mathbb{F}$ and $n \in \mathbb{N}$, define the tensor test language*

$$\mathrm{TENSORQ}(\mathbb{F}, n) := \big\{(a, b) \in \mathbb{F}^n \times \mathbb{F}^{n^2} \mid b = \mathrm{flat}(a \otimes a)\big\}.$$

Theorem 3.3.3. *For every finite field $\mathbb{F}$ and $n \in \mathbb{N}$,*

$$\mathrm{TENSORQ}(\mathbb{F}, n) \in \mathrm{LPCP}\Bigg[\varepsilon_c = 0,\ \varepsilon_s = \frac{2|\mathbb{F}| - 1}{|\mathbb{F}|^2},\ \Sigma = \mathbb{F},\ \ell = n + n^2,\ q = 3,\ r = 2n \cdot \log(|\mathbb{F}|)\Bigg].$$

Proof. Consider the following LPCP verifier $V^{\langle a, \cdot\rangle, \langle b, \cdot\rangle}$ proceeds as follows:

1. Sample $s, t \leftarrow \mathbb{F}^n$.
2. Query a at s and at t, obtaining $\langle a, s\rangle$ and $\langle a, t\rangle$.
3. Query b at $\mathrm{flat}(s \otimes t)$, obtaining $\langle b, \mathrm{flat}(s \otimes t)\rangle$.
4. Check if $\langle b, \mathrm{flat}(s \otimes t)\rangle = \langle a, s\rangle \cdot \langle a, t\rangle$.

Completeness. Suppose $b = \mathrm{flat}(a \otimes a)$. Then $\forall s, t \in \mathbb{F}^n$,

$$\begin{aligned}\langle b,\, \mathrm{flat}(s \otimes t)\rangle = \langle \mathrm{flat}(a \otimes a),\, \mathrm{flat}(s \otimes t)\rangle &= \sum_{i,j \in [n]} a_i a_j s_i t_j \\ &= \Big(\sum_{i \in [n]} a_i s_i\Big) \cdot \Big(\sum_{j \in [n]} a_j t_j\Big) = \langle a, s\rangle \cdot \langle a, t\rangle,\end{aligned}$$

Soundness. Suppose $b \neq \mathrm{flat}(a \otimes a)$, i.e., there exists $i^*, j^* \in [n]$ such that $b_{i^* j^*} \neq a_{i^*} \cdot a_{j^*}$. For each $i \in [n]$ define the linear polynomial $p_i(t) := \sum_{j \in [n]} (b_{ij} - a_i a_j)\, t_j$. The check can be rewritten as

$$\langle b, \mathrm{flat}(s \otimes t)\rangle - \langle a, s\rangle\langle a, t\rangle = \sum_{i \in [n]} \sum_{j \in [n]} (b_{ij} - a_i a_j)\, s_i t_j = \sum_{i \in [n]} p_i(t)\, s_i.$$

Immediate application of Polynomial Identity Testing yields a lower bound of $1 - \frac{2}{|\mathbb{F}|}$ but we can do better. Since $b_{i^* j^*} \neq a_{i^*} a_{j^*}$, the polynomial p_{i^*} is nonzero, so by the Schwartz–Zippel lemma applied to t,

$$\Pr_t[p_{i^*}(t) \neq 0] \geq 1 - \tfrac{1}{|\mathbb{F}|}.$$

Conditioned on $p_{i^*}(t) \neq 0$, the expression $\sum_i p_i(t)\, s_i$ is a nonzero linear polynomial in s, so by the Schwartz–Zippel lemma applied to s,

$$\Pr_{s \leftarrow \mathbb{F}^n}\Big[\sum_i p_i(t)\, s_i \neq 0 \,\Big|\, p_{i^*}(t) \neq 0\Big] \geq 1 - \tfrac{1}{|\mathbb{F}|}.$$

Hence

$$\Pr_{s,t}[\text{verifier rejects}] \;\geq\; \left(1-\tfrac{1}{|\mathbb{F}|}\right)^2,$$

and therefore

$$\Pr_{s,t}[\text{verifier accepts}] \;\leq\; 1-\left(1-\tfrac{1}{|\mathbb{F}|}\right)^2 = \frac{2|\mathbb{F}|-1}{|\mathbb{F}|^2}. \qquad \square$$

Corollary 3.3.4.

$$\text{TENSORQ}(\mathbb{F}_2, n) \in \text{LPCP}\left[\varepsilon_c = 0,\ \varepsilon_s = \frac{3}{4},\ \Sigma = \mathbb{F}_2,\ \ell = n + n^2,\ q = 3,\ r = 2 \cdot n\right].$$

Proof. Immediate. $\square$

3.4 Linear PCP to PCP

Lemma 3.4.1.

$\text{LPCP}[\varepsilon_c, \varepsilon_s, \Sigma = \mathbb{F}_2, \ell, q, r] \subseteq \text{PCP}\ [\varepsilon_c,\ \varepsilon_s' = \max\{7/8, \varepsilon_s + 1/100\}, \Sigma' = \mathbb{F}_2,\ \ell' = 2^\ell,\ q' = 3 + 2\lceil \log 100q \rceil q,\ r' = r +$

Proof. Let $L \in \text{LPCP}[\varepsilon_c, \varepsilon_s, \Sigma = \mathbb{F}_2, \ell, q, r]$, and let V be the LPCP verifier for L. Set $t = 3\lceil \log 100q \rceil$. We define a PCP verifier as follows, for any $\bar{pi} \in \mathbb{F}_2^{\mathbb{F}_2^\ell}$ and $x \in \{0,1\}^*$.

Verifier $\bar{V}^{\bar{\pi}}(x)$:

1. If $V_{\text{BLR}}^{\bar{\pi}} = 0$ output 0 and halt, otherwise proceed with the next step.
2. Independently for each $i \in [t]$, sample $r_i \leftarrow \mathbb{F}_2^\ell$.
3. Simulate V by answering each linear query a with $\text{plurality}(\bar{\pi}(a + r_i) - \bar{\pi}(r_i))_{i\in[t]}$.

Claim 3.4.2 (Completeness)**.** *For every $x \in L$, there exists $\bar{\pi} \in \mathbb{F}_2^{\mathbb{F}_2^\ell}$ such that* $\Pr[\bar{V}^{\bar{\pi}}(x) = 1] \geq 1 - \varepsilon_c$.

Proof. Since $x \in L$ and $L \in \text{LPCP}[\varepsilon_c, \varepsilon_s, \Sigma = \mathbb{F}_2, \ell, q, r]$, let $\pi \in \mathbb{F}_2^\ell$ such that $\Pr[V^{\langle \pi, \cdot \rangle}(x) = 1] \geq 1 - \varepsilon_c$. Then we construct $\bar{\pi} \in \mathbb{F}_2^{\mathbb{F}_2^\ell}$ as follows: we define $\bar{\pi}(x) = \langle \pi, x \rangle$ for each $x \in \mathbb{F}^\ell$. Using the completeness of V_{BLR} we know that $V_{\text{BLR}}^{\bar{\pi}} = 1$ with probability 1. Also, $\text{plurality}(\bar{\pi}(a + r_i) - \bar{\pi}(r_i))_{i\in[t]} = \langle \pi, a \rangle$ for any query a and any r_i. We then conclude that $V^{\langle \pi, \cdot \rangle}(x) = \bar{V}^{\bar{\pi}}(x)$ for any $r_1, \dots, r_t$. Thus $\Pr[\bar{V}^{\bar{\pi}}(x) = 1] = \Pr[V^{\langle \pi, \cdot \rangle}(x) = 1] \geq 1 - \varepsilon_c$. $\square$

Claim 3.4.3 (Soundness)**.** *For every $x \notin L$ and every $\hat{\pi} \in \mathbb{F}_2^{\mathbb{F}_2^\ell}$, one has* $\Pr[\bar{V}^{\hat{\pi}}(x) = 1] \leq \max\{7/8, \varepsilon_s + 1/100\}$.

Proof. Let $x \notin L$ and $\hat{\pi} \in \mathbb{F}_2^{\mathbb{F}_2^\ell}$. We consider two cases.

Case $\delta(\hat{\pi}, \mathbf{LIN}) \geq 1/8$. Using the soundness of V_{BLR}, we know that $\Pr[V_{\text{BLR}}^{\bar{\pi}} = 0] \geq 1/8$. Since $\Pr[\bar{V}^{\hat{\pi}}(x) = 1] \leq \Pr[V_{\text{BLR}}^{\bar{\pi}} = 1]$ we conclude $\Pr[\bar{V}^{\hat{\pi}}(x) = 1] \leq 1 - 1/8 = 7/8$.

Case $\delta(\hat{\pi}, \mathbf{LIN}) < 1/8$ Let $S = \{\tilde{\pi} \in \mathcal{LIN} : \delta(\hat{\pi}, \tilde{\pi} \leq \delta(\hat{\pi}, \mathrm{LIN}))\}$. Suppose for a contradiction that $|S| \geq 2$, and let $\bar{\pi}, \tilde{\pi} \in S$ be distinct: since $\delta(\hat{\pi}, \bar{\pi} \leq \delta(\hat{\pi}, \mathrm{LIN})) < 1/8$ and $\delta(\hat{\pi}, \tilde{\pi} \leq \delta(\hat{\pi}, \mathrm{LIN})) < 1/8$, by triangle inequality we have $\delta(\bar{\pi}, \tilde{\pi}) < 1/4$; since $|\mathbb{F}_2| = 2$, this contradicts the fact that any two distinct $\bar{\pi}, \tilde{\pi} \in \mathcal{LIN}$ satisfy $\delta(\bar{\pi}, \tilde{\pi}) \geq 1/2$.

Let then $\tilde{\pi}$ be the unique element of $\mathcal{LIN}$ with minimal distance to $\hat{\pi}$, and let $z \in \mathbb{F}^{\ell}$ the unique vector such that $\tilde{\pi}(x) = \langle z, x\rangle$ for all x. If for every query a that V asks we have $\mathrm{plurality}(\hat{\pi}(a+r_i) - \hat{\pi}(r_i))_{i\in[t]} = \tilde{\pi}(a)$, then $\bar{V}^{\hat{\pi}}(x) = V^{\langle z,\cdot\rangle}(x)$, so provided that the r_i satisfy this condition we have that $\Pr[\bar{V}^{\hat{\pi}}(x) = 1] = \Pr[V^{\langle z,\cdot\rangle}(x) = 1] \leq \varepsilon_s$. We are only left to bound the probability that for each of the q queries that V asks we have $\mathrm{plurality}(\hat{\pi}(a+r_i) - \hat{\pi}(r_i))_{i\in[t]} = \tilde{\pi}(a)$. Note that for any a and r_i, if $\hat{\pi}(a+r_i) = \tilde{\pi}(a+r_i)$ and $\hat{\pi}(r_i) = \tilde{\pi}(r_i)$ then $\hat{\pi}(a+r_i) - \hat{\pi}(r_i) = \tilde{\pi}(a)$. Hence, the proabability over one r_i that $\hat{\pi}(a+r_i) - \hat{\pi}(r_i) \neq \tilde{\pi}(a)$ is at most $2 \cdot \delta(\hat{\pi}, \tilde{\pi}) \leq 1/4$ by a union bound. Also note that $\mathrm{plurality}(\hat{\pi}(a+r_i) - \hat{\pi}(r_i))_{i\in[t]} \neq \tilde{\pi}(a)$ means that $\hat{\pi}(a+r_i) - \hat{\pi}(r_i)$ for at least $\lceil t/2 \rceil \geq t/2$ of the i's. Since the expected number of such i's is at most $t/4$, by a Chernoff bound the probability over the choice of $r_1, \dots, r_t$ that there are at least $t/2$ of such i's is at most $\exp(-t/3)$. Union bounding over the q queries that V asks, we have that the local correction fails with probability at most $q \cdot \exp(-3t) \leq 1/100$ by our choice of t. ☐

☐

Theorem 3.4.4.

$$\mathrm{QESAT}(\mathbb{F}_2, n) \in \mathrm{LPCP}[\varepsilon_c = 0, \varepsilon_s = 3/4, \Sigma = \mathbb{F}_2, \ell = n + n^2, q = 4, r = |x| + 2 * n].$$

Proof. Let $(p_1, \dots, p_m)$ be an instance of QESAT($\mathbb{F}$) with n variables.
The LPCP verifier expects a proof $\pi = (a, b) \in \mathbb{F}^{n+n^2}$ and works as follows:

Verifier $V^{(a,b)}(p_1, \dots, p_m)$:

1. Sample $r \leftarrow \mathbb{F}^m$ and $s, t \leftarrow \mathbb{F}^n$.
2. Define
$$M := \begin{bmatrix} \mathrm{coeff}(p_1) \\ \vdots \\ \mathrm{coeff}(p_m) \end{bmatrix} \in \mathbb{F}^{m\times(n+n^2)} \quad \text{and} \quad c := \begin{bmatrix} -\mathrm{const}(p_1) \\ \vdots \\ -\mathrm{const}(p_m) \end{bmatrix} \in \mathbb{F}^m.$$
(Note: coeff(p_i) := non-constant coeffs of p_i, and const(p_i) := constant coeff of p_i)
3. *(linear equation test)* $\to$ Query (a, b) at $M^T r$ and check that $\langle a \parallel b, M^T r\rangle = \langle c, r\rangle$.
4. *(tensor test)* $\to$ Query b at $\mathrm{flat}(s \otimes t)$, a at s and t, and check that
$$\langle b, \mathrm{flat}(s \otimes t)\rangle = \langle a, s\rangle \cdot \langle a, t\rangle.$$

Completeness: Suppose $p_1(a) = \dots = p_m(a) = 0$ and set $b := \mathrm{flat}(a \otimes a)$.
Then
$$\Pr_{s,t}[\langle b, \mathrm{flat}(s \otimes t)\rangle = \langle a, s\rangle \cdot \langle a, t\rangle] = 1$$
and
$$\Pr_r[\langle a \parallel b, M^T r\rangle = \langle c, r\rangle] = 1 \quad \left(\text{since } M \begin{bmatrix} a \\ \mathrm{flat}(a \otimes a) \end{bmatrix} = c\right).$$

Soundness: Suppose $\forall a \in \mathbb{F}^n\ \exists i \in [m]$ s.t. $p_i(a) \neq 0$. Fix any $\pi = (a, b)$.
Either:

(i) $b \neq \text{flat}(a \otimes a) \implies$ tensor test passes w.p. $\leq \frac{2|\mathbb{F}|-1}{|\mathbb{F}|^2}$

OR (ii) $b = \text{flat}(a \otimes a)$ and $M \begin{bmatrix} a \\ b \end{bmatrix} \neq c \implies$ linear equation test passes w.p. $\leq \frac{1}{|\mathbb{F}|}$

$\square$